

INTRODUCTION

Energy demand is continuously increasing due to rapid urbanization, population growth, and the expansion of electronic devices in everyday life. At the same time, a large amount of energy is wasted in the form of unused mechanical vibrations and human motion. One promising solution to address this challenge is **kinetic energy harvesting**, which involves converting mechanical energy from the environment into usable electrical energy.

One of the most effective methods of kinetic energy harvesting is the use of **piezoelectric transducers**. These materials have the unique property of generating electrical charge when subjected to mechanical stress such as pressure, vibration, or bending. When integrated into surfaces like floors, walkways, or tiles, piezoelectric sensors can capture energy from human footsteps and convert it into electrical power.

This project focuses on **harvesting energy from human footsteps using piezoelectric transducers** embedded in a walking surface. Each step applies mechanical force on the piezoelectric elements, producing an electrical output. This generated energy is then conditioned, stored in a power storage unit such as a rechargeable battery or supercapacitor, and later used for low-power applications like lighting, sensors, or small electronic devices.

The concept not only promotes renewable and sustainable energy generation but also demonstrates an innovative way of utilizing everyday human activity as a source of clean energy. Such systems can be implemented in high-footfall areas like railway stations, shopping malls, and educational institutions to supplement conventional power sources and improve energy efficiency.

Overall, this project aims to design and develop an efficient piezoelectric-based energy harvesting system that converts wasted mechanical energy into usable electrical energy for practical applications

2.OBJECTIVE OF THE PROJECT

The main objective of Kinetic Energy Harvesting from Human Footsteps Using Piezoelectric Transducers is to design and develop a system that converts mechanical energy generated from human walking into usable electrical energy. The system aims to efficiently capture, convert, store, and utilize this energy for practical applications.

The specific objectives are:

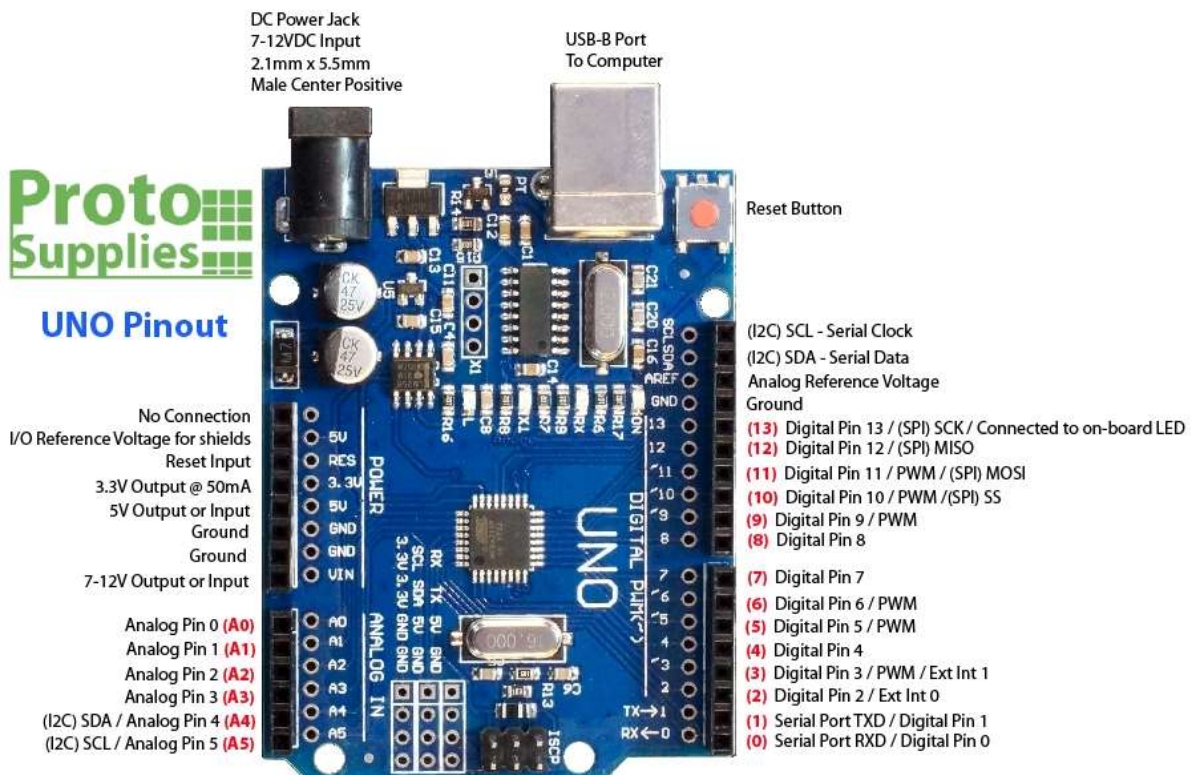
- To utilize piezoelectric transducers for converting mechanical pressure from human footsteps into electrical energy.
- To design and develop a floor-based or platform-based energy harvesting system.
- To efficiently capture kinetic energy produced by human movement in high footfall areas.
- To store the generated electrical energy in a battery or supercapacitor for future use.
- To design a suitable power conditioning circuit for regulating and stabilizing the output voltage.
- To demonstrate the utilization of stored energy for powering low-power devices such as LEDs, sensors, or small electronic circuits.
- To promote awareness and application of renewable energy sources derived from human activity.
- To reduce dependency on conventional power systems in public and commercial spaces.

3.COMPONENTS USED

COMPONENTS LIST

1. Arduino UNO R3 SMD
2. Piezo Electric Sensor
3. LCD Display
4. Jumper Wires
5. Rechargeable Battery

3.1 Arduino UNO R3 SMD



Red numbers in paranthesis are the name to use when referencing that pin.
Analog pins are references as A0 thru A5 even when using as digital I/O

Arduino UNO R3 SMD PINOUT

The Arduino Uno R3 SMD is a widely used microcontroller development board based on the ATmega328P. It is designed for easy prototyping and is commonly used in embedded systems, IoT, and sensor-based projects.

The “SMD” version differs from the standard UNO mainly in the packaging of the microcontroller—the ATmega328P is soldered directly onto the board (Surface Mounted Device), making it more compact and durable compared to the socketed (DIP) version.

Key Features

- Operating Voltage: 5V
- Input Voltage (recommended): 7–12V
- Digital I/O Pins: 14 (6 support PWM output)
- Analog Input Pins: 6 (A0–A5)
- Clock Speed: 16 MHz
- Flash Memory: 32 KB
- SRAM: 2 KB
- EEPROM: 1 KB
- USB Interface: Used for programming and power supply

Main Components

- **ATmega328 Microcontroller-** It is a single chip Microcontroller of the ATmel family. The processor code inside it is of 8-bit. It combines Memory (SRAM, EEPROM, and Flash), Analog to Digital Converter, SPI serial ports, I/O lines, registers, timer, external and internal interrupts, and oscillator.
- **ICSP pin** - The In-Circuit Serial Programming pin allows the user to program using the firmware of the Arduino board.

- **Power LED Indicator-** The ON status of LED shows the power is activated. When the power is OFF, the LED will not light up.
- **Digital I/O pins-** The digital pins have the value HIGH or LOW. The pins numbered from D0 to D13 are digital pins.
- **TX and RX LED's-** The successful flow of data is represented by the lighting of these LED's.
- **AREF-** The Analog Reference (AREF) pin is used to feed a reference voltage to the Arduino UNO board from the external power supply.
- **Reset button-** It is used to add a Reset button to the connection.
- **USB-** It allows the board to connect to the computer. It is essential for the programming of the Arduino UNO board.
- **Crystal Oscillator-** The Crystal oscillator has a frequency of 16MHz, which makes the Arduino UNO a powerful board.
- **Voltage Regulator-** The voltage regulator converts the input voltage to 5V.
- **GND-** Ground pins. The ground pin acts as a pin with zero voltage.
- **Vin-** It is the input voltage.
- **Analog Pins-** The pins numbered from A0 to A5 are analog pins. The function of Analog pins is to read the analog sensor used in the connection. It can also act as GPIO (General Purpose Input Output) pins.

Programming Arduino

Once Arduino IDE is installed on the computer, connect the board with computer using USB cable. Now open the arduino IDE and choose the correct board by selecting Tools>Boards>Arduino/Arduino Uno, and choose the correct Port by selecting Tools>Port. Arduino Uno is programmed using Arduino programming

language based on Wiring. To get it started with Arduino Uno board and blink the built-in LED, load the example code by selecting Files>Examples>Basics>Blink. Once the example code (also shown below) is loaded into your IDE, click on the ‘upload’ button given on the top bar. Once the upload is finished, you should see the Arduino’s built-in LED blinking. Below is the example code for blinking:

```
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
  delay(1000);                      // wait for a second
  digitalWrite(LED_BUILTIN, LOW);  // turn the LED off by making the voltage LOW
  delay(1000);                      // wait for a second
}
```

Arduino UNO Applications

- Weighing Machines
- Traffic Light Count Down Timer
- Parking Lot Counter
- Embedded Systems
- Home Automations
- Industrial Automation
- Medical Instrument

3.2 Piezo Electric Sensors



A sensor which works on the principle of piezoelectricity is known as a piezoelectric sensor. Where piezoelectricity is a phenomenon where [electricity is generated](#) if mechanical stress is applied to a material

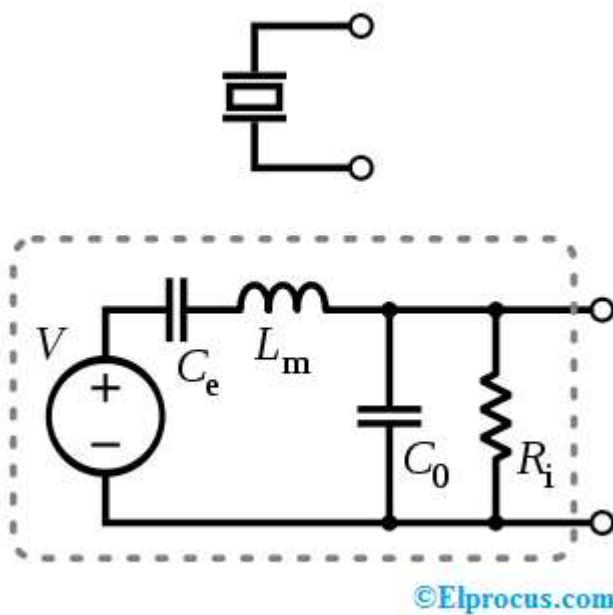
Working of a Piezoelectric Sensor

The commonly measured physical quantities by a piezoelectric sensor are Acceleration and Pressure. Both pressure and acceleration sensors work on the same principle of piezoelectricity but the main difference between them is the way force is applied to their sensing element.

In the pressure sensor, a thin membrane is placed on a massive base to transfer the applied force to the piezoelectric element. Upon application of pressure on this thin membrane, the piezoelectric material gets loaded and starts generating electrical voltages. The produced voltage is proportional to the amount of pressure applied.

In accelerometers, seismic mass is attached to the crystal element to transfer the applied force to piezoelectric materials. When motion is applied, seismic mass load's the piezoelectric material according to Newton's second law of motion. The piezoelectric material generates charge used for calibration of motion.

An acceleration compensation element is used along with a pressure sensor as these sensors can pick up unwanted vibrations and show false readings.



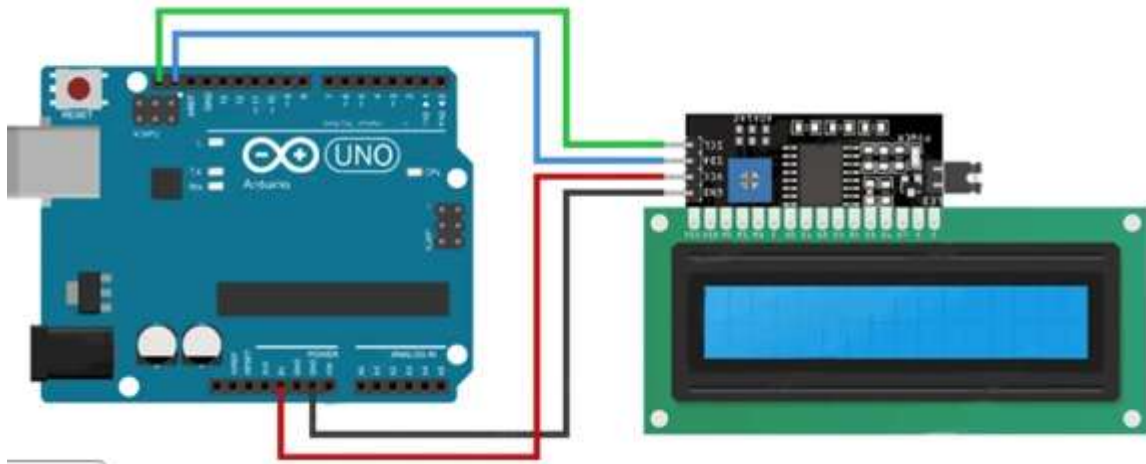
A piezoelectric sensor internal circuit is given above. The resistance R_i is the internal resistance or insulator resistance. The inductance is due to the inertia of [the sensor](#). The capacitance C_e is inversely proportional to the elasticity of the sensor material. For the proper response of the sensor, the load and leakage resistance must be large enough so that low frequencies are preserved. A sensor can be called a pressure [transducer](#) in an electrical signal. Sensors are also known as primary transducers.

3.3.LCD Display



LCD is the most commonly used electronic display module. LCD is a liquid crystal display that uses the light modulating property of liquid crystals and displays the content. A 16×2 LCD has 16 columns and 2 rows so that 16 characters can be displayed in a single row. The advantages of LCD include thin compared with other displays and the power consumption is low. LCD has various applications it is used in laptops, calculators, mobile phones etc. The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc.

LCD 16×2 Pin Diagram



I2C Interface

The I2C (Inter-Integrated Circuit) module is attached to the LCD to reduce the number of pins required for connection. Instead of using 6–8 digital pins, the I2C module uses only two communication lines:

- **SDA (Serial Data Line)**
- **SCL (Serial Clock Line)**

This makes the circuit simpler and easier to connect with microcontrollers like Arduino UNO.

3. Working Principle

The Arduino sends data to the LCD through the I2C module using serial communication. The I2C backpack converts this serial data into parallel data required by the LCD. The LCD then displays the received information such as:

- Voltage generated from piezo sensors
- Energy stored in battery
- System status (ON/OFF or charging mode)

4. Advantages of I2C LCD

- Reduces wiring complexity (only 4 connections: VCC, GND, SDA, SCL)
- Saves Arduino digital pins for other sensors

- Easy programming using libraries like LiquidCrystal_I2C
- Compact and reliable communication
- Suitable for real-time display applications

5. Application in This Project

In the piezoelectric energy harvesting system, the LCD with I2C interface is used to:

- Display real-time energy generation values
- Show system status during operation
- Provide user-friendly monitoring of output performance

Features of LCD16x2

- The operating voltage of this LCD is 4.7V-5.3V
- It includes two rows where each row can produce 16-characters.
- The utilization of current is 1mA with no backlight
- Every character can be built with a 5×8 pixel box
- The alphanumeric LCDs alphabets & numbers
- Is display can work on two modes like 4-bit & 8-bit
- These are obtainable in Blue & Green Backlight
- It displays a few custom generated characters

3.4.Jumper Wires



Jumper wires are simply wires that have connect or pin at each end allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed. Fairly simple. In fact, it doesn't get much more basic than jumper wires

A **jump wire** (also known as **jumper**, **jumper wire**, **DuPont wire**) is an electrical wire, or group of them in a cable, with a connector or pin at each end (or sometimes without them – simply "tinned"), which is normally used to interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering. Individual jump wires are fitted by inserting their "**end connectors**" into the slots provided in a breadboard, the header connector of a circuit board, or a piece of test equipment.

3.5.Rechargeable battery



A 3.9V battery is generally a lithium-ion rechargeable battery, most commonly an 18650 battery, which has a nominal voltage of 3.7V. When the battery shows a voltage of around 3.9V, it indicates that it is in a partially charged and healthy condition, since a fully charged lithium-ion battery reaches about 4.2V.

These batteries work based on the movement of lithium ions between the anode and cathode during charging and discharging, which produces electrical energy to power electronic devices. They typically have a capacity ranging from 2000 mAh to 3000 mAh, allowing them to provide power for a longer duration.

In mini projects such as a footstep power generator using piezoelectric sensors, the 3.9V battery is used to store the generated energy and supply it to components like Arduino, LCD display, or LEDs. Lithium-ion batteries are preferred due to their high energy density, lightweight nature, and rechargeability, but they require proper handling, as overcharging, deep discharging, or short-circuiting can damage the battery. Overall, a 3.9V lithium-ion battery is an efficient and reliable power source for small electronic applications.

4.HARDWARE CIRCUIT:

The circuit is designed to capture mechanical energy from footsteps using piezoelectric sensors and convert it into usable electrical energy, which is then measured and displayed.

1. Piezoelectric Sensor Array

Multiple piezoelectric sensors are arranged beneath a footstep platform. These sensors are connected in a series-parallel combination:

- Series connection increases the output voltage
- Parallel connection increases the current output

When pressure is applied (footstep), each sensor generates a small AC voltage signal due to mechanical deformation.

2. Rectifier Circuit

Since piezo sensors produce AC voltage, the output is passed through a bridge rectifier (diodes):

- Converts AC signal into pulsating DC
- Ensures current flows in one direction only
- Protects downstream components from reverse polarity

3. Filtering Stage

A capacitor is connected after the rectifier:

- Smooths the pulsating DC into a more stable voltage
- Stores small amounts of generated energy temporarily
- Reduces voltage fluctuations

4. Voltage Divider / Protection Resistor

A resistor (or voltage divider) is used:

- To limit current flowing into the Arduino
- To ensure voltage stays within safe input range (0–5V)
- Protects the analog input pin from voltage spikes

5. Arduino Interface

The conditioned DC output is connected to an analog input pin of the Arduino:

- Arduino reads the voltage level generated by footsteps
- Converts analog signal into digital values using ADC
- Processes the data for display

6. LCD Display (16x2)

The LCD is interfaced with the Arduino:

- Displays voltage or power generated
- Connected using either 4-bit or 8-bit communication mode
- Powered by Arduino's 5V supply

7. Power Supply

- Arduino is powered via USB or external 5V supply
- Piezo section is self-generating (no external supply needed)

Overall Flow

Footstep → Piezo Sensors → AC Voltage → Rectifier → Filter Capacitor → Arduino Analog Pin → LCD Display

05.SOFTWARE:

Arduino software: The Arduino software ,often referred to as the Arduino IDE(Integrated Development Environment), is a user friendly platform used for

programming Arduino microcontroller boards. Here's an overview of the Arduino software

Code Editor: The Arduino IDE provides a text editor where users can write, edit, and organize their Arduino sketches(programs).It supports syntax highlighting and auto-completion features to aid in writing code.

Compiler :The Arduino IDE includes a compiler that translates the written code into machine readable instructions for the Arduino board.

Library Manager: Arduino libraries contain pre-written code that simplifies tasks such as interfacing with sensors, controlling motors, and communicating with external devices. The Arduino IDE includes a library manager for easily installing and managing libraries.

Serial Monitor: The Arduino IDE features a built-in serial monitor that allows users to communicate with the Arduino board via serial communication. This is useful for debugging and monitoring the behavior of the Arduino sketch.

Upload Tool: The Arduino IDE enables users to upload their compiled sketches to Arduino boards via USB or other supported communication interfaces.

Board Manager: Arduino supports a wide range of microcontroller boards, each with its own specifications and capabilities. The board manager in the Arduino IDE facilitates the selection and configuration of the appropriate board for the project.

Integrated Help: The Arduino IDE includes documentation and examples to assist users in learning about Arduino programming and troubleshooting issues.

Installation: The Arduino software is available for multiple operating systems, including Windows, macOS ,and Linux. Users can download the Arduino IDE from the official Arduino website and install it on their computer following the provided instructions.

<https://www.arduino.cc/en/software>

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Bring Your Projects to Life with Arduino Software



Arduino IDE 2.3.8
Release notes

The new major release of the Arduino IDE is faster and even more powerful in addition to a more modern editor and a more responsive interface it features auto-completion, code navigation, and even a new debugger. For more details, check the [Arduino IDE 2.0 Documentation](#).

Windows (64-bit) or macOS (64-bit) **DOWNLOAD**

Nightly Builds
Download a preview of the upcoming release with the most updated features and bugfixes.

The Arduino IDE 2.0 is open source and its source code is stored on [GitHub](#).

Legacy IDE (1.8.19)
Download a legacy version of the Arduino IDE.



5.1. Program Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// LCD setup
LiquidCrystal_I2C lcd(0x27, 16, 2);
// Variables
int sensorPin = A0;
int stepCount = 0;
int threshold = 50;    // Adjust based on testing
bool triggered = false; // Prevent multiple counts
float vout;
void setup() {
    Serial.begin(9600);
    pinMode(8, OUTPUT);
    lcd.init();
    lcd.backlight();
    lcd.setCursor(0, 0);
    lcd.print("STEP COUNT:");
    lcd.setCursor(0, 1);
    lcd.print("VOLTAGE:");
}
void loop() {
    int v = analogRead(sensorPin);
    // Convert to voltage (0–5V range)
    vout = (v * 5.0) / 1023.0;
    // -----
    // STEP DETECTION (FIXED LOGIC)
    // -----
    if (v > threshold && !triggered) {
        stepCount++;
    }
}
```

```

digitalWrite(8, HIGH);
lcd.setCursor(12, 0);
lcd.print(" "); // clear old value
lcd.setCursor(12, 0);
lcd.print(stepCount);
triggered = true;
}
// Reset trigger when signal drops
if (v < threshold) {
    triggered = false;
    digitalWrite(8, LOW);
}
// DISPLAY VOLTAGE
lcd.setCursor(9, 1);
lcd.print(" "); // clear previous value
lcd.setCursor(9, 1);
lcd.print(vout * 1000); // mV
lcd.print("mV");
// SERIAL DEBUG
Serial.print("Raw: ");
Serial.print(v);
Serial.print(" | Voltage: ");
Serial.print(vout * 1000);
Serial.print(" mV | Steps: ");
Serial.println(stepCount);
delay(20); // small delay (NOT 200)
}

```

Code Explanation:

The code is designed to implement a step counting system using an Arduino, an analog sensor, and an I2C LCD display. It begins by including the necessary libraries, `Wire.h` for I2C communication and `LiquidCrystal_I2C.h` for controlling

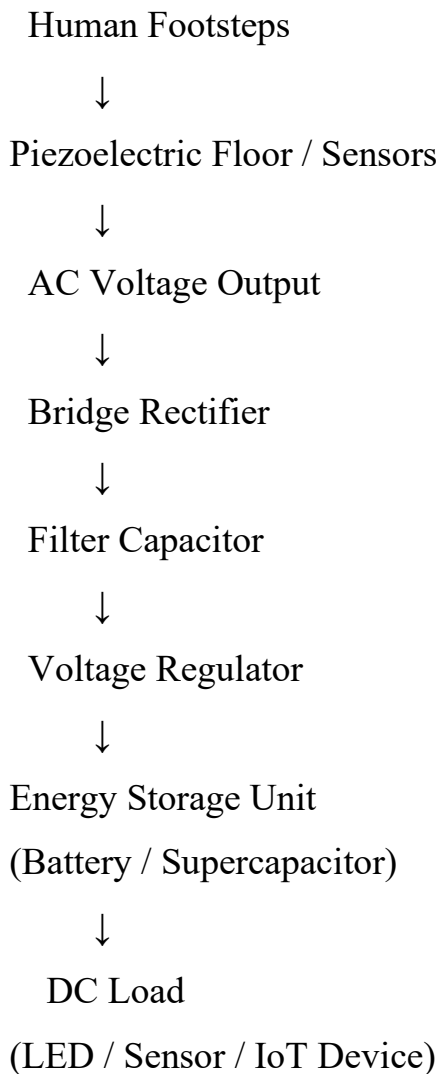
the LCD. The LCD is initialized with address 0x27 and configured as a 16×2 display.

Several variables are declared, including the sensor pin (A0), a step counter, a threshold value for detecting steps, a boolean variable called triggered to avoid multiple counts, and a variable to store the output voltage. In the setup function, serial communication is started, the LED pin is set as output, and the LCD is initialized. Labels for step count and voltage are displayed.

In the loop function, the analog value from the sensor is read and converted into voltage. If the sensor value exceeds the threshold and the system is not already triggered, the step count is increased, and the LED is turned on. The updated count is displayed on the LCD. When the value drops below the threshold, the trigger resets and the LED turns off.

The voltage is continuously displayed in millivolts, and all data is also printed to the serial monitor for debugging

5.2.Algorithm



Explanation

1. Human Footsteps

The system starts with human walking or stepping on a specially designed floor. This creates mechanical pressure.

2. Piezoelectric Sensors

Piezoelectric materials convert mechanical stress (pressure from footsteps) into electrical energy in the form of AC voltage.

3. Bridge Rectifier

Since piezoelectric output is AC, a bridge rectifier converts it into DC voltage so it can be stored.

4. Filter Capacitor

The output from rectifier is not smooth. A capacitor is used to remove ripples and stabilize the DC output.

5. Voltage Regulator

This ensures the voltage remains constant and safe for storage and load usage.

6. Energy Storage Unit

The regulated DC power is stored in a **rechargeable battery or supercapacitor** for later use.

7. Load

The stored energy is used to power small electronic devices such as LED lights, sensors, or IoT modules.

Working Principle of the Project:

The footstep power generator works based on the Piezoelectric Effect, where mechanical energy from human footsteps is converted into electrical energy. When a person steps on the platform, the piezoelectric sensors placed under it experience pressure and generate a small electrical voltage.

This generated voltage is in the form of AC, which is then converted into DC using diodes so that it can be used by electronic components. The output voltage is sent to the Arduino Uno, which reads the signal and detects each step based on a set threshold value. Whenever a valid step is detected, the system increases the step count and displays it along with the generated voltage on a 16x2 LCD Display. This process continues for every step, allowing the system to both generate and monitor energy in real time.

STEPS:-**1.Generation of Electrical Energy**

When a person steps on the platform, pressure is applied to the piezoelectric discs.

These discs (made of piezoelectric material) convert mechanical stress → electrical energy.

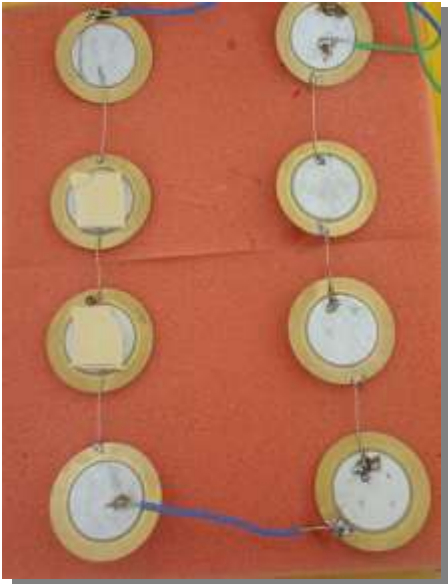
Each step produces a small AC voltage signal.

2. Combining Sensor Outputs

Multiple piezo sensors are connected together.

This increases the total generated voltage/current.

The combined output is still AC and unstable.



3. Rectification (AC to DC Conversion) And Stabilization

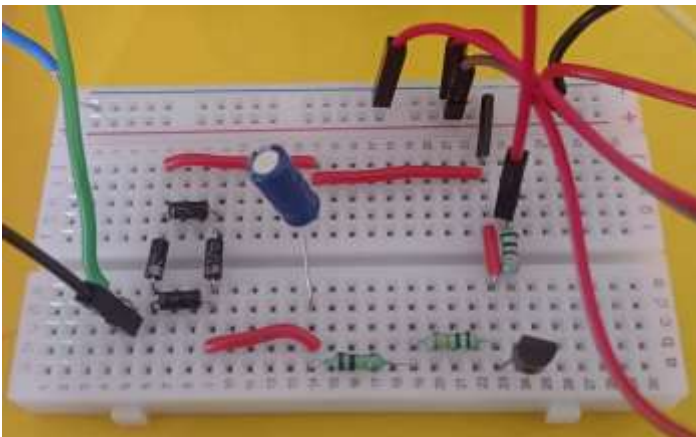
The output from piezo sensors is passed through a bridge rectifier (diodes).

This converts AC voltage into DC voltage.

The DC output becomes usable for electronic circuits.

A capacitor (if used in your circuit) smooths the fluctuating voltage.

This reduces noise and gives a more stable DC output.

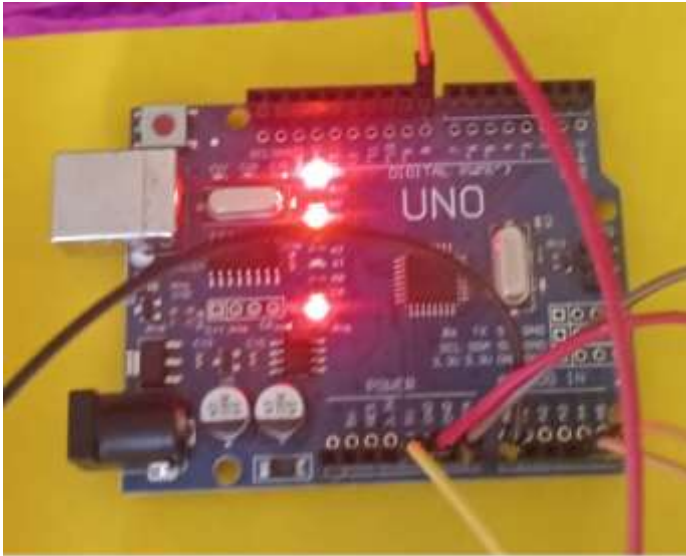


5. Voltage Measurement using Arduino

The DC voltage is given to an analog pin (A0) of the Arduino.

The Arduino Uno reads this voltage as an analog value (0–1023).

It converts this value into actual voltage using:



6. Step Detection Logic

A threshold value is set in the program.

When voltage exceeds this threshold:

It is considered a valid footprint.

Step count is incremented.

A trigger condition ensures:

One step is counted only once (avoids multiple counts for one press).

7. Display System

The calculated voltage and step count are displayed on a 16x2 LCD Display.

Example:

Line 1 → Step Count

Line 2 → Voltage generated (in mV)

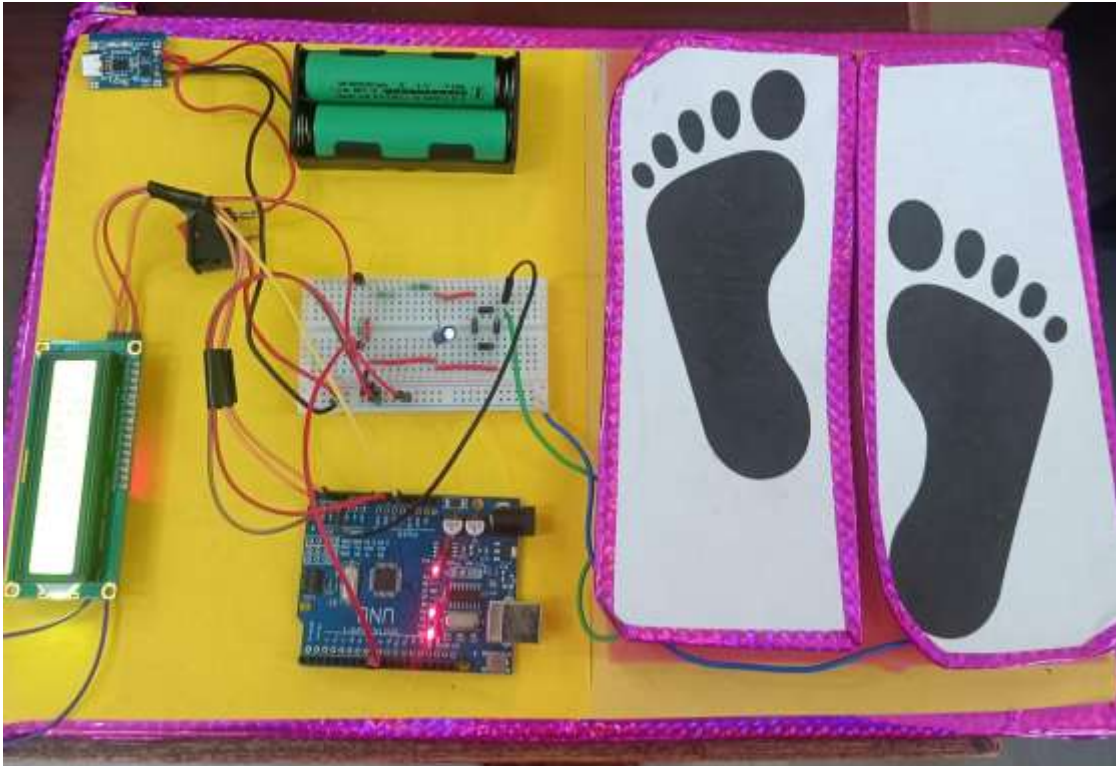


8. Continuous Operation

The system continuously monitors the piezo sensors.

Each new step generates energy and updates the display in real time.

Result and Discussion



Result

The project “Kinetic Energy Harvesting from Human Footsteps Using Piezoelectric Transducers” successfully demonstrates the conversion of mechanical energy from human footsteps into usable electrical energy. When pressure is applied on the piezoelectric sensors, electrical voltage is generated, which is then rectified, filtered, and regulated to obtain a stable DC output.

The generated energy is effectively stored in a battery or supercapacitor and can be used to power low-power electronic devices such as LEDs and sensors. The system proves that human walking activity can be utilized as a renewable source of energy, especially in high footfall areas.

Real-world Applications:

This concept can be applied in crowded areas like sidewalks, railway stations, and malls to generate small amounts of renewable energy.

Conclusion

The developed system for **Kinetic Energy Harvesting from Human Footsteps Using Piezoelectric Transducers** presents a forward-thinking approach to transforming everyday human movement into a meaningful source of electrical energy. Instead of allowing mechanical energy from footsteps to be wasted, the system intelligently captures, converts, and stores it in a usable form.

This project highlights the idea that even small, repetitive actions such as walking can collectively become a **distributed micro-energy network** when implemented on a large scale. By integrating piezoelectric materials into walking surfaces, public infrastructure can evolve into an active energy-generating platform.

Although the energy produced per step is limited, the innovation lies in scalability and sustainability. In high-footfall environments, this technology has the potential to supplement conventional power systems, reduce energy consumption for low-power applications, and support the development of **self-powered smart infrastructure**.

In conclusion, this project is not just an energy-harvesting system but a step toward redefining how cities interact with human activity—turning movement into electricity and everyday spaces into silent contributors to a greener and smarter future.

Demonstration With Class Staffs:



